From: Scherer, L.A. (Laura)
To: Elin Röös; francesca.verones@ntnu.no
Subject: Re: Potentially disappearing species from your recent publication
Date: den 24 mars 2024 21:26:17
Attachments: image001.png
image002.png
ecoregion_country_areas.csv
species_numbers.csv

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Hi Elin,

Thanks for your interest in our data.

1) You can find attached our species numbers per ecoregion. For vertebrates, they are based on IUCN data as in Chaudhary and Brooks (2018) and, therefore, similar, but they are based on a more recent version, and our values are, of course, consistent with our calculations. For plants, we derived them from a different data source, as we still needed to know which species occur where. In Chaudhary and Brooks' (2018) simpler approach, it was sufficient for them just to work with the species richness values, which they directly took from Kier et al. (2005). The values for vertebrates and our estimates for plants are an underestimate because it is only known for some species where they occur. Some ecoregions even have 0 species of a certain species group overlapping with them, for which we could then not calculate CFs with the original approach and needed to calculate proxies. The estimates from Kier et al. (2005) are much larger and closer to the actual number of species, although the number of known species increases over time. So, you might still want to use their data for an estimate of the absolute number of species lost.

2) I've recalculated and exported the areas of ecoregions over countries for you. Please note that a few countries are missing, and we used neighbouring countries as a proxy, as described in the manuscript.

Best regards,
Laura

From: Elin Röös <elin.roos@slu.se>
Sent: Friday, March 22, 2024 11:41 AM
To: Scherer, L.A. (Laura) <l.a.scherer@cml.leidenuniv.nl>; francesca.verones@ntnu.no <francesca.verones@ntnu.no>
Subject: Potentially disappearing species from your recent publication

Hi Laura and Francesca,

Thanks for your great publication providing updated characterization factors for biodiversity!

We have previously used the Chaudhary and Brooks (2018) in many applications and would now like to update to using your factors.

However, we would like to also use the absolute number of species potentially affected on a country level. I guess I could calculate these if I knew 1) the number of species for each taxa in each ecoregion and 2) the share of ecoregions per country.

For 1) this is available in Chaudhary's supplementary:

Table S2: Overview of ecoregions, habitat types, species richness and vulnerability scores of each taxa per ecoregion

ECOREGION_CODE	BIOME	REALM	ECOREGION_NAME	Habitat type	Total Species Richness (Sorg) of each taxa				
					Sorg_mammals	Sorg_birds	Sorg_amphibians	Sorg_reptiles	Sorg_plants
AA0101	1	AA	Admiralty Islands lowland rain forests	island	20	64	7	29	2000
AA0102	1	AA	Banda Sea Islands moist deciduous forests	island	19	225	2	53	1200
AA0103	1	AA	Biak-Numfoor rain forests	island	27	107	3	60	1500
AA0104	1	AA	Buru rain forests	island	24	178	5	24	1700
AA0105	1	AA	Central Range montane rain forests	island	139	348	127	79	7500
AA0106	1	AA	Halmahera rain forests	island	50	223	15	60	2000

For 2) this is something you use in your calculations.

So I wonder if that is something you potentially have in a sharable format? Potentially, you might even have the absolute CF?

Sorry, if I missed if this is already provided in the Supplementary.

Thanks in advance!

Elin

Sveriges lantbruksuniversitet
Swedish University of Agricultural Sciences

Institutionen för energi och teknik | Department of Energy and Technology
Postadress: Box 7032, 750 07 Uppsala | Postal address: P.O. Box 7032, SE-750 07 Uppsala, SWEDEN
Besöks- och leveransadress: Lennart Hjelms väg 9 | Visiting and Delivery address: Lennart Hjelms väg 9
Telefon: 018-67 18 66 | Phone: +46 18 67 18 66
Mobil: 070-305 7710 | Mobile: +46 70 305 7710
elin.roos@slu.se
Read about my reseach here:
https://www.slu.se/cv/elin-roos/
https://blogg.slu.se/new-legume-foods/
http://blogg.slu.se/food-system-group/
Read and discuss about food systems here: https://www.tabledebates.org/

När du skickar e-post till SLU så innebär detta att SLU behandlar dina personuppgifter. För att läsa mer om hur detta går till, klicka här
E-mailing SLU will result in SLU processing your personal data. For more information on how this is done, click here